2a. Overview and Goals

Martin Alfaro
PhD in Economics

> **Remark**
>
> Throughout the book, I made some deliberate choices regarding whether and when to introduce certain subjects. Considering this, I'll include a section called "Overview and Goals" prior to each chapter, which elucidates my rationale for these choices. The goal is to contextualize the book's approach, offering readers some guidance on the best way to engage with the material.

The current chapter deals with **variables and types**. Variables in Julia act as labels of objects, which in turn hold values associated with a specific type. The type determines the information the computer needs to manipulate those values. Their definitions and general properties are presented in Section 2b.

We then explore several types of objects, distinguishing between single-element collections (**scalars**) and multiple-element objects (**collections**). Numbers are a primary example of the former, including integers and floating-point values. They're introduced in Section 2c. In this group, we also discuss characters, which are scalars representing individual symbols. They lead to the coverage of **strings** in Section 2d, which are a sequence of characters that together represent text.

For collections, we focus on two of the most widely used constructs in Julia: **vectors** (presented in Section 2e) and **tuples** (discussed in Section 2f). These structures allow us to group multiple values into a single object, enabling more complex data organization and manipulation.

At this early stage, **we only scratch the surface of the topics**. In particular, the chapter doesn't cover any object in depth, and even excludes important ones. The reason is pedagogical: I didn't want to overwhelm readers with details about objects or types, considering that core programmatic concepts like functions and for-loops haven't yet been introduced.

In light of this, Chapter 2 should be understood as an introduction to objects, providing enough background to progress into the basics of working programmatically. The main skills you should gain from this chapter are:

- Familiarizing yourself with Julia's syntax
- Distinguishing between scalars (single-element objects) and collections.